

2025-2026 M.Sc. in Data Science and Analytics

Using Big Data Analytics to Improve Customer Experience and Targeting in the Banking Sector

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Objective: To determine how big data analytics and machine learning can improve customer targeting and experience in retail banking. Six classifiers are built and compared on the UCI Bank Marketing Dataset — 45,211 telemarketing records from a Portuguese bank — across three experiments: a baseline on imbalanced data, the same models with SMOTE, and hyperparameter tuning of the strongest three. The study introduces XGBoost, never previously benchmarked on this dataset, quantifies the effect of class-imbalance handling, and turns the findings into targeting recommendations for practitioners.

Background & Problem

Banks generate huge volumes of customer data through transactions, digital channels and marketing campaigns, yet direct telemarketing remains one of the least efficient channels in retail banking: conversion is low, operating costs are high, and many calls reach customers with no interest in the product.

Prior work

Moro, Cortez and Rita (2014) applied four classifiers to this dataset and reported a best AUC of 0.80 with a neural network, reaching 79% of subscribers by contacting the top half of ranked clients. Tran, Le and Nguyen (2023) compared five classifiers on a credit-card churn dataset, where Random Forest led at 97.25% accuracy.

The research gap

Neither tested XGBoost, now the standard for structured tabular data, and neither applied class-imbalance handling — although 88.3% of records are non-subscribers, so a model can look accurate while missing most subscribers.

Research Questions

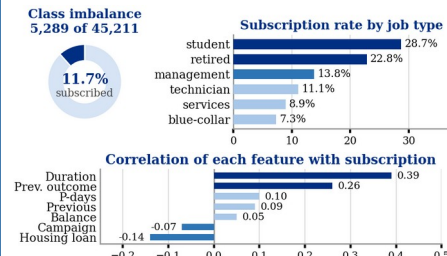
1. Which classifier gives the highest predictive performance?
2. How do results compare with published benchmarks?
3. What is the impact of SMOTE on model performance?
4. Which characteristics most strongly predict subscription?

Contributions

- First benchmark of XGBoost on this dataset.
- First systematic evaluation of SMOTE across six classifiers here.
- Linear models gain most from SMOTE; ensembles are already partly robust.
- Consensus feature ranking, plus evidence that excessive contact reduces uptake.

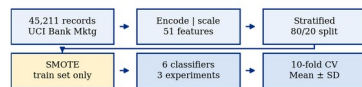
The Data

45,211 contact records from a Portuguese bank, described by 17 variables in four groups: demographics (age, job, marital, education), financial status (default, balance, housing, loan), campaign information (contact, day, month, duration, campaign, pdays, previous, poutcome) and the binary target. No values are missing.



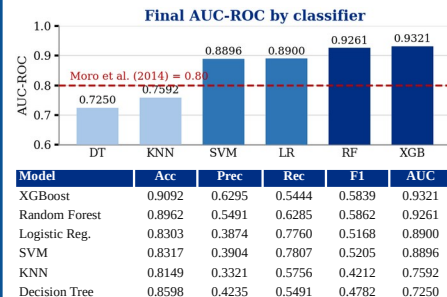
Methodology & Experimental Design

- **Preprocessing:** duplicates removed; binary, ordinal and one-hot encoding (17 → 51 features); Min-Max scaling.
- **Validation:** Mann-Whitney U and Chi-square tests confirm feature-target relationships.
- **Split & SMOTE:** stratified 80/20 split; SMOTE (k = 5) on the training set only — no synthetic data in the test set; seed 42.
- **Experiments:** (1) baseline, (2) SMOTE, (3) GridSearchCV tuning of the top three models.



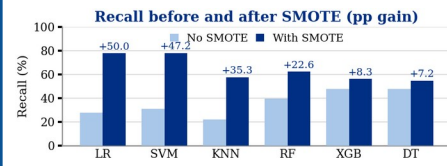
Model Performance

- **XGBoost leads all three experiments** — AUC 0.9299 at baseline, 0.9287 with SMOTE and 0.9321 after tuning, with a cross-validation SD of 0.0005.



Impact of SMOTE

Recall improves for every classifier. Logistic Regression rises from 27.9% to 77.9% and SVM from 30.9% to 78.1%. Accuracy falls (LR 89.9% → 83.1%), but the models go from finding a quarter of subscribers to more than three-quarters — the right trade when a missed subscriber costs more than a wasted call.



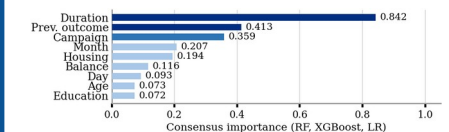
Benchmarking vs Published Work

XGBoost's 0.9321 exceeds Moro et al.'s best by 0.13, and even Logistic Regression here outperforms their neural network. Evaluation protocols differ — they used a rolling-window scheme — so the comparison is indicative rather than controlled.

Model	Moro et al. (2014)	This study	Δ
Logistic Regression	0.75	0.8900	+0.14
Decision Tree	0.73	0.7250	-0.01
SVM	0.78	0.8896	+0.11
Neural Network (best)	0.80	not tested	—
XGBoost (new)	not tested	0.9321	+0.13

Key Predictors

Consensus of Random Forest importance, XGBoost gain and LR coefficients. Duration leads but is only observable after a call; previous outcome and number of contacts are actionable beforehand. The LR coefficient for contacts is negative (−7.237).



Conclusions & Recommendations

- Use XGBoost with SMOTE; without imbalance handling the best model misses over half of all subscribers.
- Prioritise prior-campaign responders, favour cellular contact, cap repeat calls.
- Duration is post-call: retrain without it, test ADASYN, and audit fairness before deployment.

Code and results: github.com/samuelukoha/Bank-Customer-Targeting